Software Engineering

Tutorial Five (Agile)

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Agile Methods

- Rapid software development and delivery is often the most important requirement for software systems due to the fast change of the business.
- In rapid software development and delivery process; specification, design, and implementation are interleaved so the system is developed as a series of versions with stakeholders involved in each version evaluation.
- Dissatisfaction with the overheads involved in Software design led to the creation of agile methods.
- The aim of the agile methods is to reduce this overhead in the software process and to be able to respond quickly to changing requirements without excessive rework.
- Agile methods:
  - Are based on Iterative approach of software development.
  - Are intended to deliver working software quickly and evolve this to meet changing reuirements.
  - Focus on the code rather than the design.

Agile Principles

Principle	Description
Customers involvement	They should be closely involved in the development process and their role is to : ▪ Provide an prioritize the new system requirements. ▪ Evaluate the iterations.
Incremental delivery	The software is developed in increments according to the requirements which the customer specifies to be in each increment.
People not process	The skills of the development team members should be recognized and exploited, and the team members should be left to develop their own way without perspective processes.
Embrace change	Expect the system requirements to change and so the design system to accommodate these changes.
Maintain simplicity	Focus on simplicity of both the software being developed and the development process.

Agile Manifesto and its four core values:

- The agile manifesto is a document that outlines a set of guiding principles for adaptive and customer-centric software development. It is a set of foundational principles for agile software development that emphasizes flexibility, collaboration, and customer satisfaction.
- The primary purpose of agile manifesto is to provide a foundation for developing software in a way that responds effectively to changing requirements and delivers value to customers.
- The **Four core values** :
  - Individuals and Interactions over Processes and Tools:
    - Focuses on the importance of effective communication and collaboration among team members.
  - Working Software over Comprehensive Documentation:
    - Prioritizes the delivery of functional software as the primary measure of progress.
  - Customer Collaboration over Contract Negotiation:
    - Encourages customers and stakeholders to have active involvement throughout the development process.
  - Responding to Change over Following a Plan:
    - On changing requirements, embracing flexibility and ability to adapt even late in the development process.

Agile common problems

1. It can be difficult to keep the interest of customers who are involved in the process.
2. Team members may be unsuited to the intense involvement that characterizes agile methods.
3. Prioritizing changes can be difficult where there are multiple stakeholders.
4. Maintaining simplicity requires extra work.
5. Contracts may be a problem as with other approaches to iterative development.

Agile method applicability

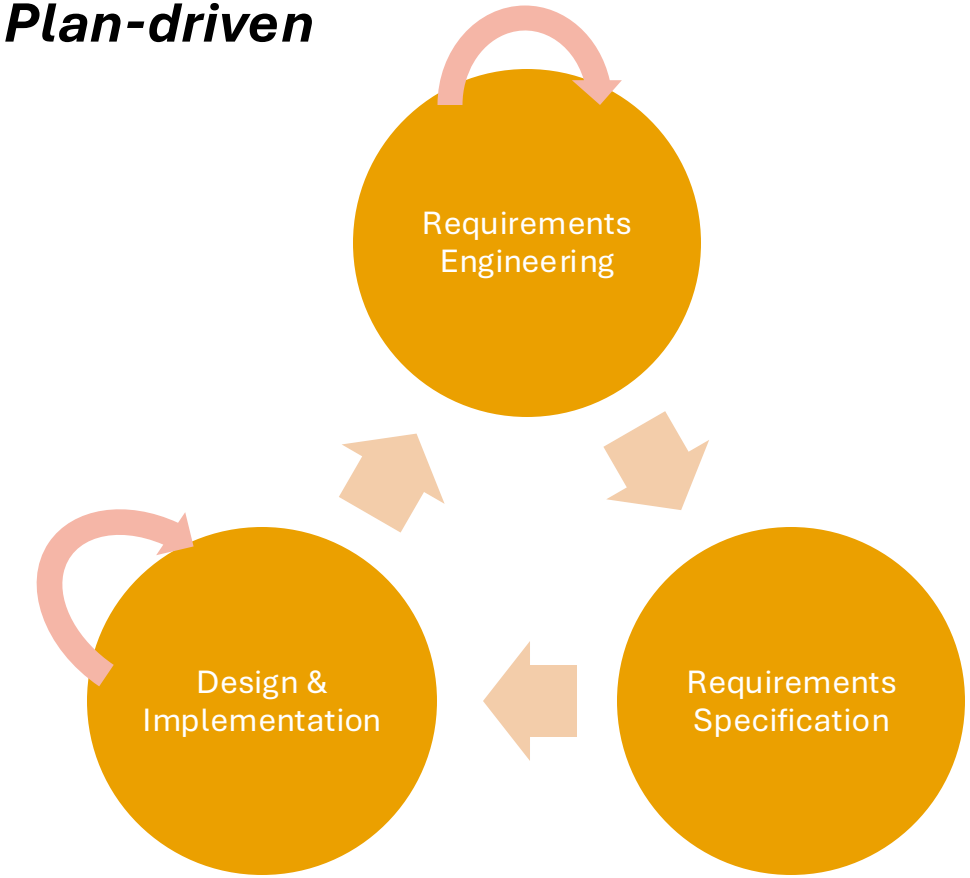
Product development where a software company is developing a small or medium-sized product for sale.

Custom system development within an organization, where there is a clear commitment from the customer to become involved in the development process and where there are not a lot of external rules and regulations that affect the software.

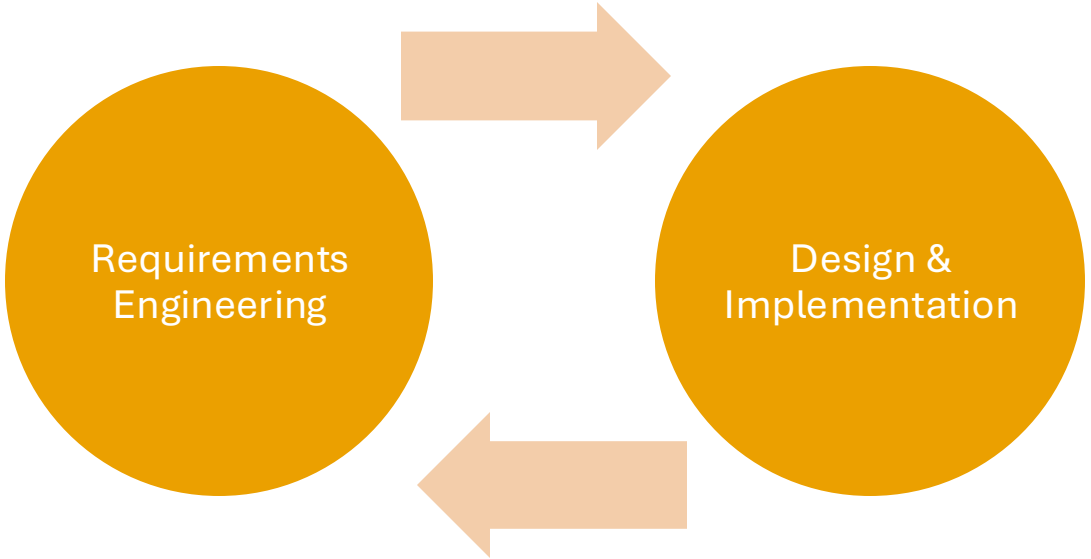
Because of their focus on small, tightly-integrated teams, there are problems in scaling agile methods to large systems.

Plan-driven Dev. VS Agile Dev.

Plan-driven



Agile



Extreme Programming (XP)

It is the best-known and most widely used agile method.

XP takes an "Extreme" approach to iterative development.

- New Versions may be built several time per day.
- Increments are delivered to customers every two weeks.
- All tests must be run for every build and the build is only accepted if tests run successfully.

Customer Role in XP

In XP, a customer or user is part of the XP team and is responsible for making decisions on requirements.

User requirements are expressed as scenarios or user stories.

These are written on cards and the development team break them down into implementation tasks. These tasks are the basis of schedule and cost estimates.

The customer chooses the stories for inclusion in the next release based on their priorities and the schedule estimates.

Refactoring in XP and its importance

- **<u>What is Refactoring ?</u>**
  - Refactoring is the process of improving the internal structure of existing code without changing its external behavior.
  - It involves making small, incremental changes to the code to enhance its readability, maintainability, and overall design.
  - In XP, refactoring is not a one-time event; it's a continuous activity performed throughout the development lifecycle.

- **<u>Why it is important ?</u>**
  - Improved Code Quality:
    - It helps to eliminate code duplication, simplify complex logic, and clarify unclear code.
  - Enhanced Maintainability:
    - Well-structured code is easier to understand and modify, reducing the effort required for future maintenance and updates.
  - Increased Flexibility:
    - makes it easier to adapt to changing requirements and incorporate new features, which is essential in the agile environment of XP.
  - Support for Continuous Integration:
    - Because XP heavily utilizes continuous integration, refactoring must be done in small steps, with frequent testing, so that the code remains stable.

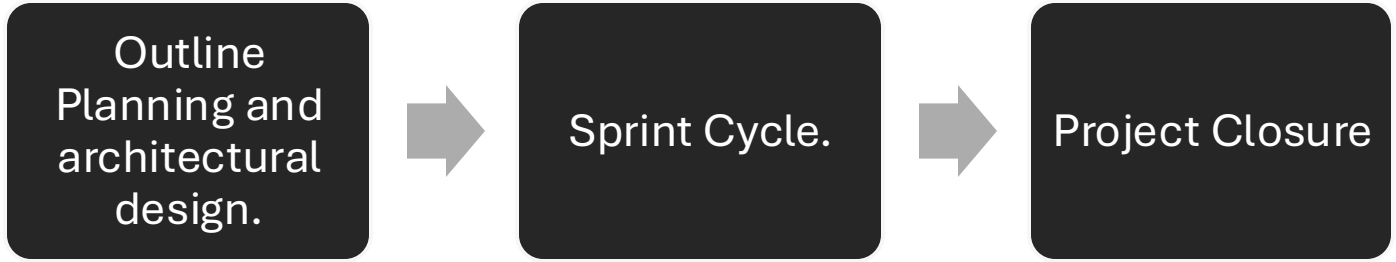
Agile Project Management

- The principal responsibility of software project managers is to manage the project so that the software is delivered on time and within the planned budget for the project.
- The standard approach to project management is plan driven. Managers draw up a plan for the project showing what should be delivered, when it should be delivered and who will work on the development of the project deliverables.
- Agile project management requires a different approach, which is adapted to incremental development and the particular strengths of agile methods. Which is **"The Scrum Approach"**.

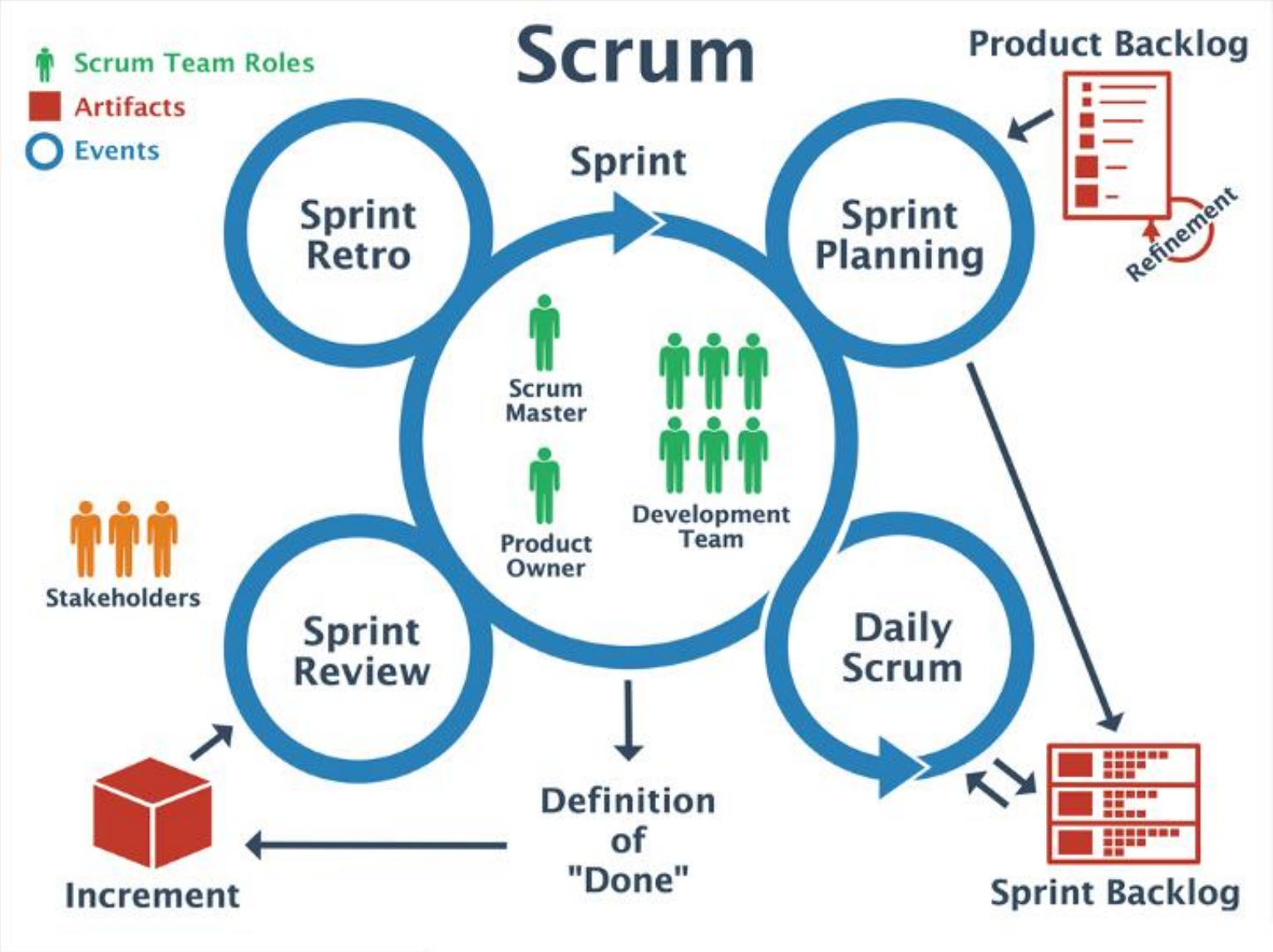
Scrum Methodology

- The Scrum approach is a general agile method, but its focus is on managing iterative development rather than specific agile practices.
- **Scrum Phases :**
    1) The initial phase: is an outline planning phase where you establish the general objectives for the project and design the software architecture.
    2) This is followed by a series of sprint cycles, where each cycle develops an increment of the system.
    3) The project closure phase wraps up the project, completes required documentation such as system help frames and user manuals and assesses the lessons learned from the project.

Scrum Process



Sprint Cycle



The role of the Scrum master is to protect the development team from external distractions.

Challenges of scaling Agile methods to large software systems

1. Coordination and Communication:
   - **Increased Team Size:** Large projects involve numerous teams, making communication and coordination significantly more complex.
   - **Geographical Distribution:** Distributed teams across different time zones introduce further communication barriers, requiring sophisticated tools and processes.
2. Architectural and Technical Challenges:
   - **Integration Complexity:** Integrating numerous components developed by different teams can be challenging.
   - **Architectural Consistency:** Maintaining a consistent architecture across numerous teams and components is crucial. This requires strong architectural governance and clear guidelines.
3. Planning and Management Challenges:
   - **Release Planning:** Coordinating releases across multiple teams and components requires meticulous planning and synchronization.
4. Organizational and Cultural Challenges:
   - **Skill Gaps:** Teams may lack the necessary skills and experience to work effectively in a scaled Agile environment.